END-TO-END SYSTEM OF MBTI PERSONALITY RECOGNITION

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Abstract

Myers–Briggs Type Indicator (MBTI) is one of the most widely used frameworks for personality classification. this framework have broad applications, including therapy, education and talent acquisition. Despite its widespread use, accurately identifying an individual's personality type through self-report assessments is challenging due to subjectivity, bias, and limited reliability. To address these limitations, this project aims to develop a machine learning based approach for personality recognition by analyzing linguistic patterns within textual data. The project focuses on building an end-to-end system for personality classification based on the MBTI framework. The system will involve data collection, preprocessing, feature extraction, model training, and evaluation, ultimately enabling automated prediction of personality types from user-generated text. By leveraging advances in natural language processing and machine learning, this work seeks to provide a more scalable and data driven solution for personality assessment.

Introduction

MBTI Personality Types 500 Dataset is a dataset that contains about 106K records of preprocessed 500-word posts and their authors' self-reported personality types[1][4][7]. To increase further this dataset, we cleaned and lemmatized another similar one called MBTI Personality Type Twitter Dataset[3]. Then, we joined both datasets, totaling 113,878 records.

	posts	type	ie	ns	tf	jp
0	know intj tool use interaction people excuse a...	INTJ	I	N	T	J
1	rap music ehh opp yeah know valid well know fa...	INTJ	I	N	T	J
2	preferably p hd low except wew lad video p min...	INTJ	I	N	T	J
3	drink like wish could drink red wine give head...	INTJ	I	N	T	J
4	space program ah bad deal meing freelance max ...	INTJ	I	N	T	J
...	...	...	...	...	...	...
113873	godpls take care hiro emergency room be you ok...	INTP	I	N	T	P
113874	wow last time I get intp I think u upset the f...	INTP	I	N	T	P
113875	a that someone will get his ass kick so its ok...	ENTP	E	N	T	P
113876	if you re intj this one be for you what be nev...	INFJ	I	N	F	J
113877	hey can you dm I a pic of what harry be wear c...	ISTP	I	S	T	P

Fig. 1: A view of the dataset that was to train 4 binary MBTI models

Methods

The train_test_split function was used to split the dataset[6]. The parameter set was: (all_datasets['posts'], y, test_size=0.3, random_state=42, stratify=y). Then, the training set(70%) was fed into a Scikit-learn Pipeline[5] with TfidfVectorizer and LogisticRegression.

```
text_clf = Pipeline([
    ('tfidf', TfidfVectorizer(
            max_features=50000,   # max num of words accross all posts
            ngram_range=(1, 2),   # unigrams + bigrams('feel','feel like
            sublinear_tf=True,    # replace tf with 1 + log(tf) (ignores
            min_df=3,             # word must appear in at least 3 posts
    )),

    ('clf',LogisticRegression(
            class_weight='balanced',
            max_iter=1000,
            random_state=42,
            C=1.0,
    )),
])
```

Fig. 2: The vectorizer and classification algorithm used for training along their arguments.

Results

After fitting each binary model, predictions were run against the test set(34,163 stratified records) for each MBTI dimension. This means that 4 full iterations on the dataset was performed. In average, the four models reached a 89% accuracy. The dimension with the higher rate of true positives was the FT(feeling and thinking) class. On the other hand, the dimension that saw the worst performance was the JP(judging and perciving) one.

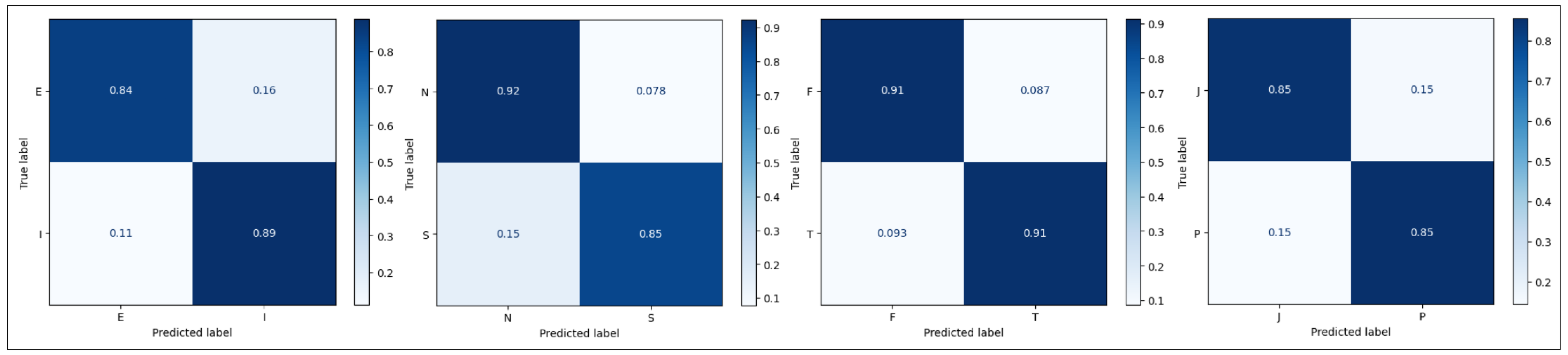


Fig. 3: The confusion matrix of each binary model normalized over the true conditions.

A web application was then developed from scratch to serve these models. It hosted them in joblib format. As users submit their speech samples via the fronted, the application process their inputs in the backend and redirects the users to the results page displaying the models' predictions.

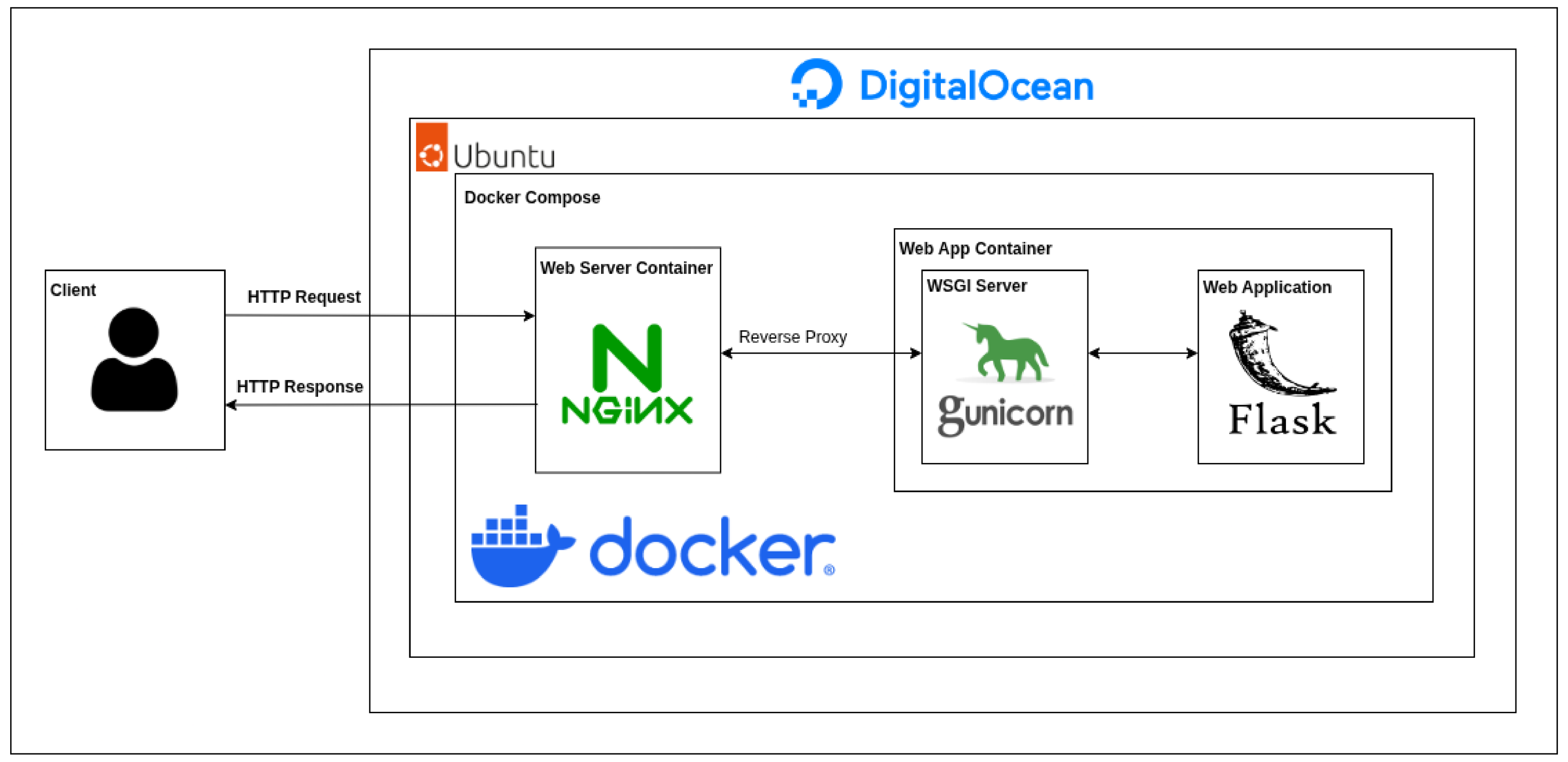
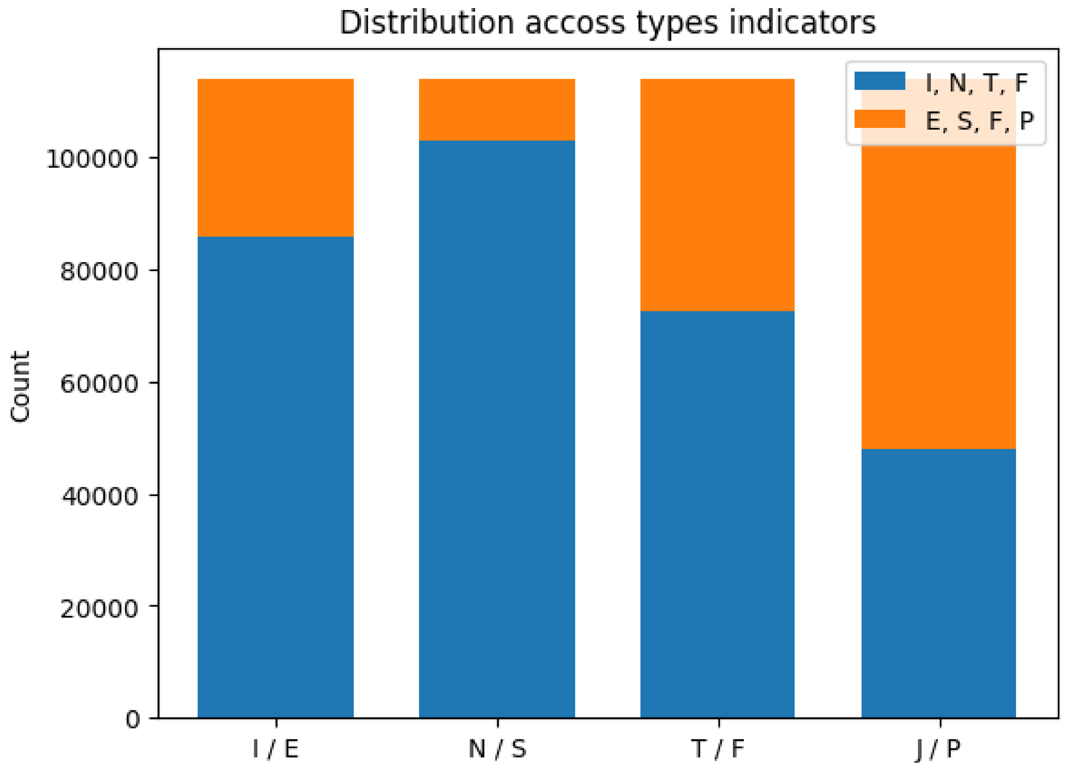
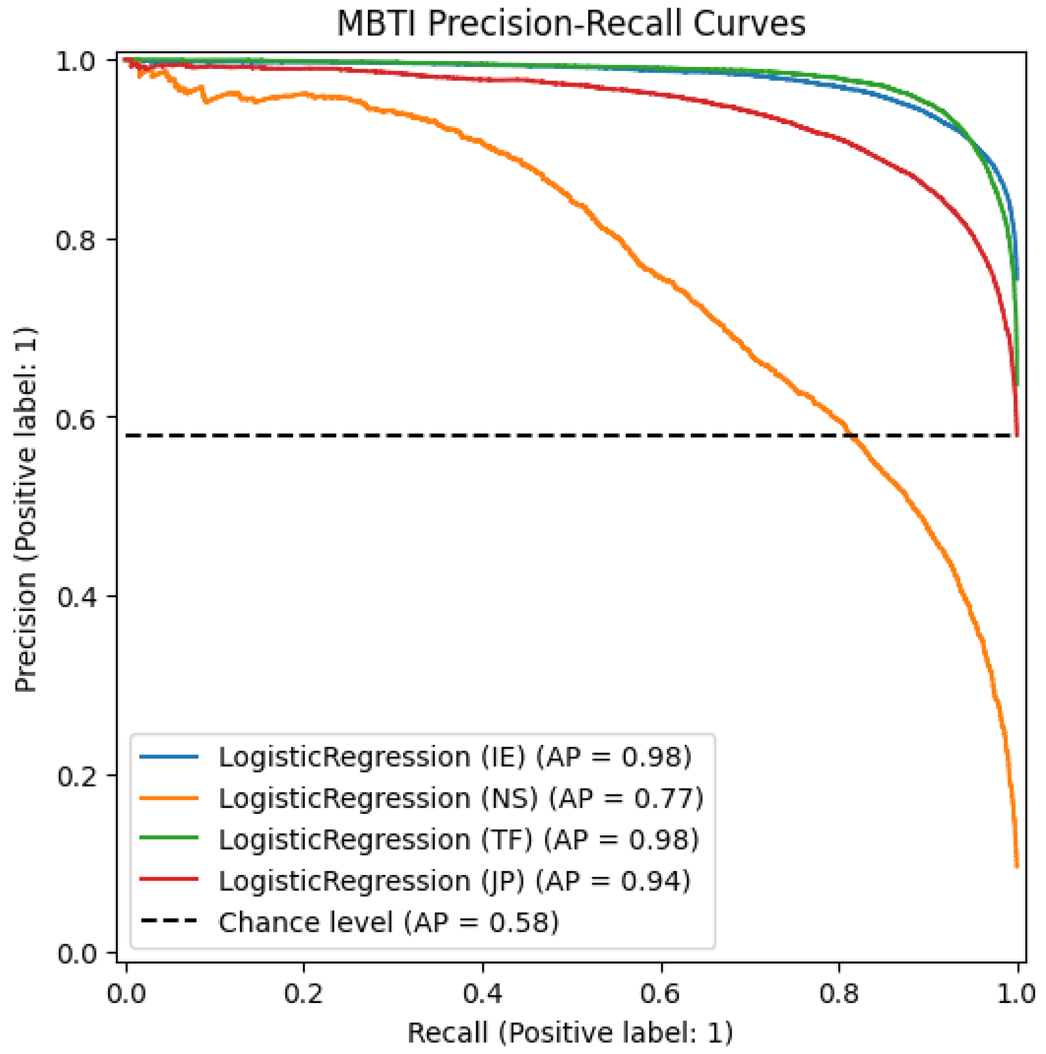


Fig. 4: End-to-End system design diagram

"The precision-recall curve shows the tradeoff between precision and recall for different thresholds. A high area under the curve represents both high recall and high precision"





Conclusions

In conclusion, an end-to-end system for MBTI personality classification was built and deployed. We utilized Google Colab Notebooks to collect 113,878 records from public MBTI datasets. We preprocessed, cleaned and lemmatized user-generated text from social media posts to increase fitting speed. We performed feature engineering by breking down MBTI labels into dimensions to train four Logistic Regression models, achieving an average of 89% accuracy across the four. Finally, we enabled automatic prediction of personality from user speech patterns by deploying a web application built with Flask, to host the pretrained models in the backend and return prediction results to users. Future work will involve exploring and integrating a Word2Vec transformer with Sckit-Learn pipelines to improve the precision of the current models.

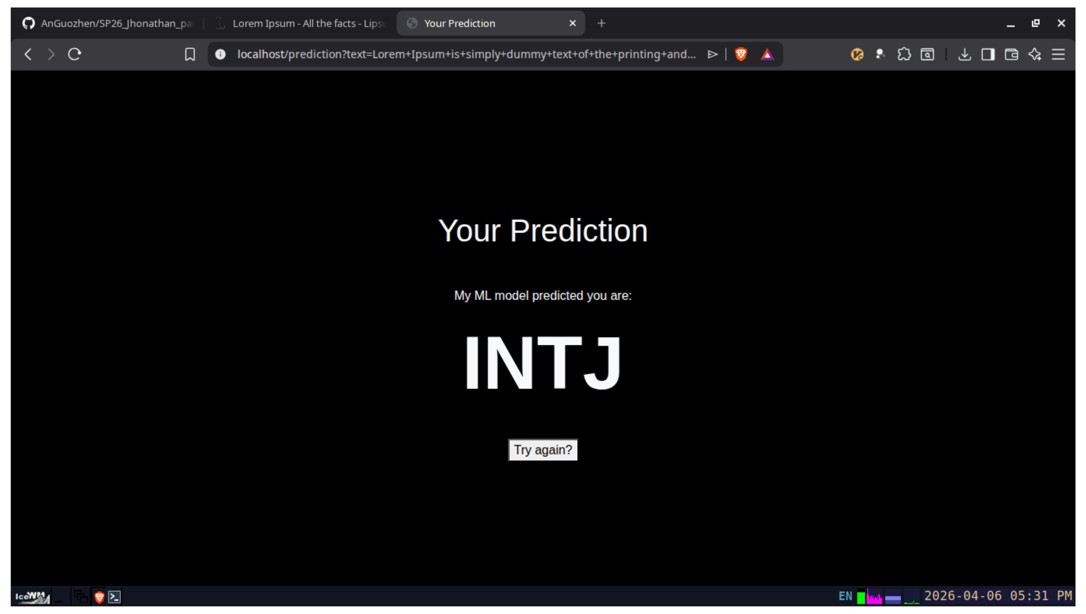


Fig. 6: Fronted of web application.

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